

Guidelines for operationalization-orientated manual annotation of ecological aspects in literary texts

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By selecting the three categories plants, animals, and habitats as main categories used for manual annotation we are following the Fauna-Flora-Habitat (FFH) classification system from the European Habitats Directive 92/43/EEC, which can be found here:

<https://eur-lex.europa.eu/legal-content/DE/TXT/?uri=CELEX:31992L0043>

The guidelines are the result of a repetitive annotation workflow following Reiter et al. (2020). For the development of the categories, we primarily worked with texts selected from [EcoCor](#), which consists of 31 ecocritically relevant texts from 1790 to 1920 (Barkey et al. 2024). They show that an annotation is not determined solely by the mere occurrence of a word; rather, the specific context of the entire passage is taken into account.

Plants

Definition: A living organism that uses energy from sunlight—and, rarely, chemical energy—to synthesize organic matter from inorganic substances.

Annotation span: Mostly one or two words.

Examples for plants in literary texts which are annotated:

- Named plants: “heather,” “rose,” “ivy,” “herbs”
- Parts of plants: “seed,” “stem,” “vine,” “crown,” “lawn”
- Mystical plants that do not exist in the real world: “tree of life”
- Rather vague references to plants or formations of plants (i.e., when they stand alone and are not depicted as a habitat): “shrubs”
- Shaped plants or formations of plants (i.e., when they stand alone and are not depicted as a habitat): “hedge”
- Artificial plants (e.g., in coats of arms, paintings, statues, and wreaths)
- Hybrids like “mushrooms”

The following aspects are not annotated:

- Plant-based products such as “wine” or “bread” if they no longer have a direct connection to plants or nature, but have instead been further processed by humans.
- Plant characteristics (like “red” in “the red rose”)
- Human or animal character names that are identical to plant names (like “Rose” in “Rose, the blacksmith’s daughter”)

Metaphors and comparisons:

- When a plant metaphor is used to describe a plant, it is annotated (e.g., “green sea” for grass). In this case, the characteristics are also annotated.
- When a plant metaphor is used to describe a human character, it is not annotated (“there stood the wallflower, lonely and far away from all the others”).
- Exception: When a plant is clearly the focus of a comparison relating to a character, it is annotated (e.g., “she was as beautiful as a rose, yet covered in thorns”) .

Animals

Definition: A living organism that is more mobile, more responsive to stimuli, and has a more complex internal structure than a plant. Unlike plants, it obtains only organic substances as food and obtains the oxygen required for its metabolism through specialized respiratory organs.

Annotation span: Mostly one or two words.

Examples for animals in literary texts which are annotated:

- Named animals: “fox,” “dog,” “lion”
- Part of living animals: “paw,” “mouse ear,” “power feathers”
- Mythical animals that do not exist in the real world: “griffin”
- Artificial animals: in paintings, coats of arms, or statues
- Non-living, unprocessed animal parts: “skeleton”

The following aspects are not annotated:

- Animal products such as “skins” or “drinking horns,” when they are no longer directly associated with the animal, but have instead been further processed by humans.
- Sounds that animals make (like the neighing of horses or the howling of wolves)
- Any products left behind by animals (like “molehills,” “tracks”)
- Features that describe animals further (like “big” in “the big elephant”)
- Human names of characters that are identical to animal names (like “Wolf” in “the Wolf family”)

Metaphors and comparisons:

- If an animal metaphor refers to a specific animal (“The gray-feathered thief” refers to “a heron”), it is annotated.
- If an animal metaphor does not refer to an animal, it is not annotated (“There came Theodor, the proud lion, to lead his army into battle”).
- If a comparison to an animal is used to further describe a human character but at the same time specifically emphasizes a feature of the animal, it is annotated (like “lion” in “He was strong like a lion”).
- If an animal's name is used as an insult but doesn't share information about the animal itself, it is not annotated (“he was a cowardly dog”)

Habitats

Definition: Areas, (natural) habitats for plants and animals; areas (a geographically defined region with a clearly delineated boundary), natural habitats (entirely natural or near-natural terrestrial or aquatic areas characterized by geographical, abiotic, and biotic features), and specific habitats of a species (a habitat determined by specific abiotic and biotic factors in which that species occurs at any stage of its life cycle).

Annotation span: Mostly one or two words.

Examples for habitats in literary texts which we annotated:

- Named habitats: "Matterhorn," "Elbe"
- Natural habitat: "heath," "forest," "forest edge," "field," "meadow," "mountain," "slope," "North Pole," "hedgerow," "garden," "pond," "ditch," "field," "fallow land," "thicket," "scrub," "shore," "nest," "cave"
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- Elements of a habitat: "The lakeshore was sandy," "the edge of the forest"
- Entities that are explicitly inhabited by animals: Bushes, hedges, or windbreaks that are inhabited by animals ("The hedgehog disappeared into the bushes," "The hedge was teeming with insects")
- Vaguely, but defined areas: "Vistula," "region," "bay," "coast," "island," "wasteland"
- Mystical realms that do not correspond to the real world: "Underworld," "Mount Olympus," "Styx," "magic cave"

The following aspects are not annotated:

- Rather open-ended, vague references to the environment, such as "ground" and "nature".
- Country names that emphasize the political entity: "France," "Poland," "Neumark".
- Larger geographical units such as "world," "earth," or "empire" are annotated only if they are not intended in a political, territorial, or ideological sense.
- Places where the human character's personal connection to that place is the focus ("home," "homeland," "country of birth")
- "World" as a moral standpoint ("That's where worlds collide")
- Potential habitats (such as "rocks" and "cliffs") are annotated only when it is clear from the specific textual context that plants or animals are present in the surrounding area and turn the abstract space into a populated habitat (like rock in "The rock was covered with lichen," or mountaintop in "A single tree could be seen on the mountaintop").
- Habitats that do not focus on non-human inhabitants, but rather on humans ("village," "city," "town").
- The air and the sky are not classified as habitats.

Metaphors and comparisons:

- Only metaphors and comparisons used to describe a natural habitat are annotated.
- Metaphors that describe non-natural aspects and center on human behavior or life are not annotated ("A little girl has been born into this world" as a metaphor for "birth")

Literature and related publications

Barkey, Sören, Ingo Börner, Clara Funk, Viktor Illmer, Carsten Milling, Mark Schwindt, Daniil Skorinkin, Henny Sluyter-Gäthje, and Peer Trilcke. *EcoCor*. URL: <https://ecocor.org/doc/credits> [Access: 26.03.2026].

Schumacher, Mareike, Marie Flüh, and Lempp, forthcoming. 'Ecological Codes: Constructing Nature in Literature', in: Book of Abstracts of the DH Conference 2025: Accessibility and Citizenship. Lissabon.

Reiter, Nils (2020). 'Anleitung zur Erstellung von Annotationsrichtlinien', in Reiter, Nils, Axel Pichler, and Jonas Kuhn (eds) *Reflektierte algorithmische Textanalyse*. Berlin: De Gruyter, pp. 193–202. DOI: <https://doi.org/10.1515/9783110693973-009>.